Trends in Inorganic Chemistry

01.	Melting point	=	Li > Na > K > Rb > Cs
02.	Colour of the flame	=	Li-Red, Na-Golden, K-Violet, Rb-Red, Cs-Blue, Ca-Brick red, Sr-Blood red, Ba-Apple green
03.	Stability of hydrides	=	LiH > NaH > KH > RbH> CsH
04.	Basic nature of hydroxides	=	LIOH < NaOH < KOH < RbOH < CsOH
05.	Hydration energy	=	Li> Na > K> Rb > Cs
06.	Reducing character	=	Li > Cs > Rb > K > Na
07.	Stability of +3 oxidation state	=	B> Al > Ga > In > T1
08.	Stability of +1 oxidation state	=	Ga < In < Tl
09.	Basic nature of the oxides and hydroxides	=	B< Al< Ga < In < Tl
10.	Relative strength of Lewis acid	=	$BF_3 < BCl_3 < BBr_3 < BI_3$
11.	Lonisation energy	=	B> Al <Ga > In <Tl
12.	Reactivity	=	C<Si< Ge < Sn <Pb
13.	Metallic character	=	C< Si < Ge < Sn < Pb
14.	Acidic character of the oxides	=	$CO_2 > SiO_2 > GeO_2 > SnO_2 > PbO_2$
15.	Reducing nature of hydrides	=	$CH_4 < SiH_4 < GeH_4 < SnH_4 < PbH_4$
16.	Thermal stability of tetrahalides	=	$CCl_4 > SiCl_4 > GeCl_4 > SnCl_4 > PbCl_4$
17.	Oxidising character of M+4 species	=	$GeCl_4 < SnCl_4 < PbCl_4$
18.	Ease of hydrolysis of tetrahalides	=	$SiCl_4 < GeCl_4 < SnCl_4 < PbCl_4$

19.	Acidic strength of trioxides	=	$N_2O_3 > P_2O_3 > As_2O_3$
20.	Acidic strength of pentoxides	=	$N_2O_2 > P_2O_2 > As_2O_2 > Sb_2O_2 > Bi_2O_2$
21.	Acidic strength of oxides of nitrogen	=	$N_2O < NO < N_2O_3 < N_2O_4 < N_2O_5$
22.	Basic nature/bond angle/Thermal stability and dipole moment of hydrides	=	$NH_3 > PH_3 > AsH_3 > SbH_3 > BiH_3$
23.	Stability of trihalides of nitrogen	=	$NF_3 > NCl_3 > NBr_3$
24.	Lewis base strength	=	$NF_3 < NCl_3 < NBr_3 < NI_3$
25.	Ease of hydrolysis of trichlorides	=	$NCl_3 > PCl_3 > AsCl_3 > SbCl_3 > BiCl_3$
26.	Lewis acid strength of trihalides of P, As, and Sb	=	$PCl_3 > ASCl_3 > SbCl_3$
27.	Lewis acid strength among phosphorus trihalides	=	$PF_3 > PCl_3 > PBr_3 > PI_3$
28.	Melting and boiling point of hydrides	=	$H_2O > H_2Te > H_2Se > H_2S$
29.	Volatility of hydrides	=	$H_2O < H_2Te < H_2Se < H_2S$
30.	Reducing nature of hydrides	=	$H_2S < H_2Se < H_2Te$
31.	Covalent character of hydrides	=	$H_2O < H_2S < H_2Se < H_2Te$
32.	The acidic character of oxides (elements in the same oxidation state)	=	$SO_2 > SeO_2 > TeO_2 > PoO_2$ $SO_3 > SeO_3 > TeO_3$
33.	Acidic character of oxide of a particular element (e.g. S)	=	$SO < SO_2 < SO_3$ $SO_2 > TeO_2 > SeO_2 > PoO_2$
34.	Bond energy of halogens	=	$Cl_2 > Br_2 > F_2 > I_2$
35.	Solubility of halogen in water	=	$F_2 > Cl_2 > Br_2 > I_2$

36.	Oxidising power	=	$F_2 > Cl_2 > Br_2 > I_2$
37.	Enthalpy of hydration of X ion	=	$F^- > Cl^- > Br^- > I^-$
38.	Reactivity of halogens	=	F> Cl> Br > I
39.	Ionic character of M-X bond in halides	=	M-F > M-Cl > MBr > M-I
40.	Reducing character of X ion	=	I- > Br- > Cl- > F-
41.	Acidic strength of halogen acids	=	HI > HBr > HCl > HF
42.	Reducing property of hydrogen halides	=	HF < HCL < HBr < HI
43.	Oxidising power of oxides of chlorine	=	$Cl_2O > ClO_2 > Cl_2O_6 > Cl_2O_7$
44.	Decreasing ionic size	=	$O^{2-} > F^- > Na^+ > Mg^{2+}$
45.	Increasing acidic property	=	$Na_2O_3 < MgO < ZnO < P_2O_5$
46.	Increasing bond length	=	$N_2 < O_2 < F_2 < CL_2$
47.	Increasing size	=	$Ca_{2+} < Cl_- < S_{2-}$
48.	Increasing acid strength	=	$HClO < HClO_2 < HClO_3 < HClO_4$
49.	Increasing oxidation number of iodine	=	$HI < I_2 < ICl < HIO_4$
50.	Increasing thermal stability	=	$HOCl < HOClO < HOClO_2 < HOClO_3$